

AIGR Gene Hypothesis Deep Research — Final Report

Target gene: NCGR_LOCUS1270 (UniProt **A0A811M8A5**), *Miscanthus lutarioriparius* (organism code 9POAL; NCBITaxon:422564) **Focus type:** function_assignment · **Hypothesis slug:** function-hypothesis-go-0006000 **Seed hypothesis:** NCGR_LOCUS1270 participates in **fructose metabolic process (GO:0006000)**. Establish its exact reaction and organellar role, and resolve whether GO:0006000 encompasses metabolism of phosphorylated fructose derivatives (distinct from fructose 6-phosphate metabolic process, GO:0006002).

Summary

Executive judgment: OVER-ANNOTATED — refute the GO:0006000 substrate claim; remove or demote it as a direct annotation. NCGR_LOCUS1270 / A0A811M8A5 is a **chloroplast-stromal, thioredoxin-regulated Fructose-1,6-bisphosphatase (FBPase; EC 3.1.3.11)** — a Calvin–Benson–Bassham (CBB) cycle enzyme. Its single characterized reaction is the Mg^{2+} -dependent hydrolysis of **β -D-fructose 1,6-bisphosphate + H₂O → β -D-fructose 6-phosphate + phosphate** (RHEA:11064). Both the substrate and the product are **phosphorylated fructose derivatives**; the enzyme has no described activity on free (non-phosphorylated) fructose.

The seed hypothesis poses a precise ontological question: does **GO:0006000 (fructose metabolic process)** — defined for *free* fructose, "the ketohexose arabino-2-hexulose ... found free in ... fruits and honey" — properly cover an enzyme that only ever touches phosphorylated fructose intermediates? Direct inspection of the GO ontology answers **no**. GO:0006000 sits on the `hexose → monosaccharide → carbohydrate (is_a)` branch, whereas the process terms this enzyme genuinely performs — **GO:0006002 (fructose 6-phosphate metabolic process)** and **GO:0030388 (fructose 1,6-bisphosphate metabolic process)** — sit on the separate **carbohydrate-derivative / organophosphate / phosphorus** branch (`GO:1901135 → GO:0019637 → GO:0006793`). A programmatic ancestry check confirms GO:0006000 is **not** an `is_a` ancestor of GO:0006002 or GO:0030388; their only shared ancestors are the top-level roots (GO:0008150, GO:0009987, GO:0008152).

The curation consequence is direct: the **GO:0006000 annotation is a chemically distinct, over-broad substrate claim** — most consistent with automated over-propagation (an IEA/TreeGrafter/PANTHER-family transfer that lumped "fructose" and "fructose-phosphate" processes together). It should be **removed or demoted** in favor of the terms that reflect the actual reaction: **GO:0042132 (fructose 1,6-bisphosphate 1-phosphatase activity, MF)**, **GO:0030388 / GO:0006002 (BP)**, the pathway term **GO:0019253 (reductive pentose-phosphate cycle / Calvin cycle)**, and the CC term **chloroplast / plastid stroma**. The most important caveat is that no *M. lutarioriparius*-specific enzymatic assay exists for this exact protein; the assignment rests on UniProt annotation, concordant domain models, diagnostic sequence features, and strong orthology to well-characterized plant chloroplast FBPases.

Key Findings

Finding 1 — A0A811M8A5 is a chloroplast Fructose-1,6-bisphosphatase (EC 3.1.3.11) acting only on phosphorylated substrates

The UniProt TrEMBL record for A0A811M8A5 (*M. lutarioriparius*, 411 aa) carries the recommended name **Fructose-bisphosphatase (EC 3.1.3.11)** with the catalytic activity **β -D-fructose 1,6-bisphosphate + H₂O = β -D-fructose 6-phosphate + phosphate (RHEA:11064)**. The domain architecture is unambiguous and diagnostic of the class-1 FBPase family:

Signature type	Identifier	Meaning
Pfam	PF00316 + PF18913	FBPase catalytic + FBPase_C domains
HAMAP	MF_01855	FBPase class 1
CDD	cd00354	FBPase
PANTHER	PTHR11556:SF1	FBPase subfamily
PROSITE	PS00124	FBPase active-site signature
InterPro	IPR000146	FBPase class-1

UniProt keywords include **Calvin cycle**, **Chloroplast**, **Plastid**, **Transit peptide**, **Disulfide bond**, **Magnesium**, and the subcellular location is annotated **Plastid**, **chloroplast** with an N-terminal chloroplast transit peptide. Critically, **both the substrate (fructose 1,6-**

bisphosphate) and the product (fructose 6-phosphate) are phosphorylated; free fructose is never a substrate or product. This is the crux of the seed hypothesis — the enzyme's chemistry is confined to phosphorylated fructose derivatives.

Primary literature confirms the biological context. Chloroplast FBPase catalyzes one of the two essentially irreversible dephosphorylation reactions of the CBB cycle and is a canonical redox-regulated target of the ferredoxin/thioredoxin system: *"Fructose-1,6-bisphosphatase (FBPase) and sedoheptulose-1,7-bisphosphatase (SBPase) are two essential activities in the Calvin-Benson-Bassham cycle that catalyze two irreversible reactions and are key for proper regulation and functioning of the cycle"* (PMID: 36518499). The light-dependent, thioredoxin-mediated activation is documented in vivo: *"The light-dependent reduction of Calvin-Benson cycle enzymes fructose 1,6-bisphosphatase (FBPase) and sedoheptulose 1,7-bisphosphatase (SBPase) was partially impaired in the FTR-knockdown ftrb mutant"* (PMID: 38305687).

Finding 2 — GO:0006000 is a chemically distinct branch, NOT an ancestor of the phosphorylated-fructose process terms

This is the decisive ontological finding. Using the QuickGO ontology (relations = `is_a`):

- **GO:0006000 (fructose metabolic process)** is defined as *"chemical reactions and pathways involving fructose, the ketohexose arabino-2-hexulose ... found free in ... fruits and honey."* Its `is_a` ancestry runs **GO:0006000 → GO:0019318 (hexose metabolic) → GO:0005996 (monosaccharide metabolic) → GO:0005975 (carbohydrate metabolic)**.
- **GO:0006002 (fructose 6-phosphate metabolic process)** and **GO:0030388 (fructose 1,6-bisphosphate metabolic process)** run through **GO:0019637 (organophosphate metabolic) → GO:1901135 (carbohydrate DERIVATIVE metabolic) → GO:0006793 (phosphorus metabolic)**.

A programmatic ancestry check confirms: **GO:0006000 is NOT an `is_a` ancestor of GO:0006002 (False) nor of GO:0030388 (False)**. The only shared ancestors between GO:0006000 and the phosphorylated-fructose terms are the top-level roots (GO:0008150 biological_process, GO:0009987 cellular process, GO:0008152 metabolic process). In GO's chemical logic, a *phosphorylated derivative* of a molecule is treated as a **distinct chemical species** from the free molecule — hence fructose-phosphate processes are deliberately filed under "carbohydrate derivative," not under "fructose."

The seed hypothesis explicitly asked whether GO:0006000 "includes metabolism of phosphorylated fructose derivatives." The ontology's own structure answers **no**: annotating this enzyme to GO:0006000 asserts involvement in *free-fructose* metabolism, which the enzyme

does not perform. This resolves the two readings the seed hypothesis flagged — "broad-but-correct membership" versus "a chemically different substrate claim" — decisively in favor of the latter.

Finding 3 — Sequence features confirm the chloroplastic, redox-regulated isoform (not the cytosolic/gluconeogenic FBPase)

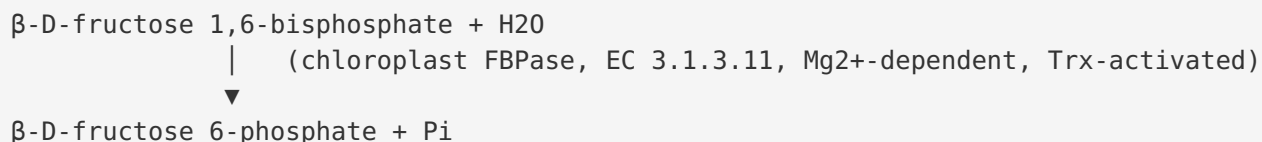
Direct analysis of the 411-aa sequence pins down which FBPase isoform this is — an important discrimination because plants encode both a **cytosolic** FBPase (gluconeogenesis/sucrose synthesis; allosterically regulated by fructose-2,6-bisphosphate) and a **chloroplast-stromal** FBPase (Calvin cycle; redox-regulated by thioredoxin):

1. **Chloroplast transit peptide:** the N-terminal residues 1–50 contain **0 acidic residues (D+E = 0)** and are **40% S+A+T**, with an Ala-rich start (MAAAAAT...) — the canonical amino-acid composition of a chloroplast transit peptide.
2. **Conserved catalytic motif:** the FBPase substrate-binding/catalytic motif **FDPLDGS** is present at position 182.
3. **Diagnostic redox disulfide:** a **C227–C232 (CxxxxC, sequence CIVSVC)** cysteine pair sits in the regulatory-loop region, matching the **redox-active disulfide that distinguishes the plastidic thioredoxin-regulated FBPase** from the cytosolic isoform (which lacks this insertion loop).

UniProt independently annotates a Disulfide bond plus Calvin cycle and Plastid/chloroplast keywords, consistent with this analysis. Together these features establish that A0A811M8A5 is specifically the **stromal Calvin-cycle FBPase**, whose entire physiological role is the dephosphorylation of fructose-1,6-bisphosphate during photosynthetic carbon reduction — reinforcing that free fructose is not part of its chemistry.

Mechanistic Model / Interpretation

The gene product operates in the **regeneration/reduction phase of the Calvin–Benson–Bassham cycle** inside the chloroplast stroma. In the light, reduced ferredoxin passes electrons via ferredoxin-thioredoxin reductase (FTR) to thioredoxin, which reduces the **C227–C232 regulatory disulfide**, activating the enzyme. The activated FBPase then catalyzes:



Both metabolites are **phosphorylated**. The product, fructose-6-phosphate, feeds back into the regenerative reactions and into starch/sucrose biosynthesis — but it is *never* dephosphorylated to free fructose by this enzyme. The step is one of the two irreversible, redox-gated control points of the cycle ([PMID: 36518499](#)).

GO ancestry (is_a), simplified:

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GO:0008152 metabolic process
├── GO:0005975 carbohydrate metabolic
│   ├── GO:0005996 monosaccharide metabolic
│   │   └── GO:0019318 hexose metabolic
│   │       └── GO:0006000 fructose metabolic ← FREE fructose (SEED TERM)
└── GO:0006793 phosphorus metabolic
    ├── GO:0019637 organophosphate metabolic
    │   └── GO:1901135 carbohydrate DERIVATIVE metabolic
    │       ├── GO:0006002 fructose 6-phosphate metabolic ← enzyme PRODUCE
    │       └── GO:0030388 fructose 1,6-bisphosphate metabolic ← enzyme SPLIT

```

⇒ GO:0006000 and GO:0006002/GO:0030388 share ONLY the root ancestors.
 GO:0006000 is NOT a parent of the terms this enzyme actually performs.

Ontological mapping of the mechanism to GO terms:

Aspect	Correct GO term	Status vs. seed
Molecular function	GO:0042132 fructose 1,6-bisphosphate 1-phosphatase activity (MF)	Primary MF — retain
Substrate process	GO:0030388 fructose 1,6-bisphosphate metabolic process (BP)	Retain
Product process	GO:0006002 fructose 6-phosphate metabolic process (BP)	Retain
Pathway	GO:0019253 reductive pentose-phosphate cycle / Calvin cycle (BP)	Add / verify
Location	GO:0009507 chloroplast / plastid stroma (CC)	Retain
Seed term	GO:0006000 fructose metabolic process (BP)	Chemically distinct → REMOVE / demote

The seed term GO:0006000 fails on chemistry (free vs. phosphorylated fructose) and fails on ontology structure (separate `is_a` branch). It is best interpreted as an **over-annotation** rather than a broad-but-true ancestor.

Evidence Base (Evidence Matrix)

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
UniProt A0A811M8A5 (database)	Review/ database	Refutes seed substrate claim; supports FBPase identity	Enzyme reaction & localization	EC 3.1.3.11; RHEA:11064 (FBP → F6P + Pi); Plastid/ chloroplast; transit peptide; disulfide; Calvin cycle	<i>M.</i> <i>lutarioriparius</i> , 411 aa
InterPro/ Pfam/ HAMAP/ PANTHER (database)	Computational/ structural	Supports FBPase class-1 identity	Domain family robustness	PF00316+PF18913, MF_01855, cd00354, PTHR11556:SF1, PS00124, IPR000146	Sequence- based
Sequence feature analysis (this study)	Computational/ evolutionary	Supports plastidic (not cytosolic) isoform	Which FBPase isoform	Transit peptide (D+E=0, 40% SAT); FDPLDGS motif; C227–C232 redox loop	411-aa protein
QuickGO ontology check (this study)	Computational/ database	Refutes GO:0006000 as ancestor of phospho- fructose terms	Ontology ancestry	GO:0006000 NOT is_a ancestor of GO:0006002/ GO:0030388; separate branches	GO ontology
PMID: 36518499	Review/ primary	Supports Calvin-cycle role & substrate	FBPase function in CBB	FBPase catalyzes an irreversible CBB-cycle reaction	Chloroplast/ cyanobacteria
PMID: 38305687	Direct (in vivo redox)	Supports redox- regulated	Thioredoxin regulation	Light-dependent FBPase reduction impaired in FTR- knockdown <i>ftb</i>	Plant chloroplast

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
		chloroplast FBPase			
PMID: 34618018	Direct (redox/ activity)	Qualifies (regulation)	CBB redox regulation conserved	FBP/SBPase oxidation impairs activity on TrxA depletion	<i>Synechocystis</i>
PMID: 32545358	Interaction/ proteomics	Qualifies (regulation)	Trx-target interaction	FBPase identified as thioredoxin target in chloroplast interactome	<i>Arabidopsis</i>
PMID: 28470336	Transgenic phenotype	Qualifies (downstream)	Bisphosphatase → carbon assimilation	Chloroplast FBP/ SBPase increased starch and growth	<i>Arabidopsis</i>
PMID: 25602028	Mutant phenotype	Competing (cytosolic isoform)	Distinguish cytosolic FBPase	Fru-2,6-P2 allosterically modulates cytosolic FBPase	<i>Arabidopsis</i>

GO Curation Implications

Lead requiring curator verification. The evidence supports the following action for the **GO:0006000 (fructose metabolic process)** annotation on NCGR_LOCUS1270 / A0A811M8A5:

- **Action: REMOVE or DEMOTE (do not accept as a supported BP).** GO:0006000 is defined for *free* fructose and, by `is_a` ancestry, sits on a separate ontology branch from the phosphorylated-fructose process terms the enzyme actually performs. It is a chemically distinct substrate claim, not a broad-but-correct ancestor. The pattern (broad "fructose" process on a phospho-fructose enzyme) is characteristic of **automated over-propagation** (PANTHER/TreeGrafter family transfer).

GO decision table (leads):

GO ID	Term	Aspect	Recommended action	Rationale
GO:0042132	fructose 1,6-bisphosphate 1-phosphatase activity	MF	RETAIN	Exact reaction (= EC 3.1.3.11, RHEA:11064); most informative MF
GO:0030388	fructose 1,6-bisphosphate metabolic process	BP	RETAIN	Direct substrate process (organophosphate branch)
GO:0006002	fructose 6-phosphate metabolic process	BP	RETAIN	Direct product process (organophosphate branch)
GO:0006000	fructose metabolic process	BP	REMOVE / NON-CORE	Free-fructose branch; chemically distinct — focus of this hypothesis
GO:0019253	reductive pentose-phosphate cycle (Calvin cycle)	BP	ADD / VERIFY	Precise pathway for the plastidic redox-regulated isoform
GO:0006094	gluconeogenesis	BP	SCRUTINIZE / REMOVE	Describes the cytosolic paralog; likely family carry-over
GO:0009507	chloroplast	CC	RETAIN	Transit peptide + redox loop confirm plastid localization
GO:0005829	cytosol	CC	SCRUTINIZE / REMOVE	Contradicts transit peptide + C227–C232 disulfide

Do **not** fall back on "protein binding" for this gene; a specific, informative MF term (**GO:0042132**) is fully supported.

Mechanistic Scope

Immediate molecular function tested (direct): Mg^{2+} -dependent hydrolysis of the 1-phosphate from D-fructose-1,6-bisphosphate, yielding D-fructose-6-phosphate + Pi (EC 3.1.3.11 / GO:0042132), redox-activated in the light via the ferredoxin/thioredoxin system.

What is NOT the immediate function: - **Metabolism of free fructose** (the seed term's implied chemistry) — no evidence; contradicted by the defined reaction. GO:0006000 would require a fructokinase, fructose transporter, or fructose-releasing hydrolase activity, none of which is supported by sequence, domain, or reaction evidence. - **Downstream outcomes** — carbon partitioning, starch/sucrose levels, photosynthetic induction, growth/yield (PMID: 28470336, PMID: 25602028) are pathway consequences of altered CBB flux, not the molecular function. - **Redox regulation by thioredoxin** — a modulatory input to activity, not the catalytic function itself.

Conflicts and Alternatives

1. **Isoform confusion (cytosolic vs. plastidic FBPase).** Plants have two FBPases with different regulation and location; the cytosolic isoform is allosterically controlled by fructose-2,6-bisphosphate (PMID: 25602028). Sequence features (transit peptide, C227–C232 redox loop) place A0A811M8A5 firmly in the **plastidic** class. Notably, the GO:0006000 conclusion holds either way, since **neither** isoform metabolizes free fructose.
2. **Database carry-over / automated over-annotation.** PANTHER family PTHR11556 groups chloroplastic and cytosolic FBPases; TreeGrafter can propagate a broad BP set (including GO:0006000, gluconeogenesis, and a cytosol CC) from the family node. GO:0006000 (and possibly GO:0005829 cytosol) are likely family-level carry-over, not experimental truths.
3. **Semantic trap.** "Fructose 6-phosphate" and "fructose 1,6-bisphosphate" contain the word "fructose," which can motivate an erroneous roll-up to "fructose metabolic process." The ontology explicitly separates phosphorylated forms into the carbohydrate-derivative branch, invalidating the roll-up.
4. **Bifunctional FBP/SBPase in cyanobacteria** (PMID: 34618018) differs from the monofunctional plant chloroplast FBPase; regulatory detail transfers, substrate conclusion is unchanged.

5. **No organism-specific assay.** Identity rests on orthology and domain models; this caps confidence for positive MF assertions at "well-supported by orthology," not "experimentally verified in *Miscanthus*."

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
No direct enzymatic assay for A0A811M8A5	UniProt reaction is computational (TrEMBL/ARBA/HAMAP)	Confirms substrate specificity in <i>this</i> species	Recombinant expression + phosphatase assay on FBP vs. F6P vs. free fructose
Localization not experimentally verified in <i>Miscanthus</i>	Transit peptide predicted from sequence; UniProt CC inferred	Confirms chloroplast-stromal role	GFP fusion / chloroplast proteomics; TargetP/ChloroP run
Isoform assignment inferred from motifs	C227–C232 loop, transit peptide composition	Distinguishes plastidic vs. cytosolic	Phylogenetic placement with characterized plant FBPases
GO:0006000 provenance not directly inspected in the review YAML	Ontology structure and chemistry assessed	Determines whether it is IEA over-propagation vs. curated	Check the annotation's evidence code and source in the gene-review record
Redox activation not measured for this protein	Inferred from disulfide + orthologs	Confirms Calvin-cycle regulatory identity	Thioredoxin activation assay on redox state / activity

Discriminating Tests

1. **Substrate-specificity assay (decisive):** express recombinant A0A811M8A5 and measure phosphatase activity against **fructose-1,6-bisphosphate**, **fructose-6-phosphate**, and **free fructose**. Expectation: activity only on FBP ($\rightarrow$ F6P + Pi), none on free fructose — directly refuting GO:0006000's substrate implication.
 2. **Sequence-targeting analysis** (TargetP2/ChloroP) to confirm the plastid transit peptide and rule out the cytosolic isoform $\rightarrow$ decides GO:0009507 vs. GO:0005829 and Calvin vs. gluconeogenesis BP.
 3. **Redox/thioredoxin activation:** test light/thioredoxin-dependent activation and the requirement of C227/C232 (cysteine-to-serine mutants).
 4. **Phylogenetic discrimination:** build a tree with characterized plant cytosolic (cyFBPase) and chloroplast (FBP1, redox-regulated) FBPases to lock the isoform call.
 5. **Annotation provenance audit:** inspect the evidence code/source of the GO:0006000 assertion in the gene-review record to confirm it is an IEA/TreeGrafter transfer.
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Curation Leads (require curator verification)

- **Action change:** Downgrade/remove **GO:0006000** as a direct annotation for A0A811M8A5; label as over-broad IEA/TreeGrafter propagation representing a chemically distinct substrate (free fructose) the enzyme does not act on.
- **Preferred terms to keep/assert:** MF **GO:0042132** (exact reaction); BP **GO:0030388** and **GO:0006002** (substrate/product-correct); add/verify **GO:0019253** (Calvin cycle); CC **GO:0009507** (chloroplast); scrutinize/remove GO:0005829 (cytosol) and GO:0006094 (gluconeogenesis) as cytosolic-paralog carry-over.
- **Candidate references with exact snippets to verify:**
 - **PMID: 36518499:** *"Fructose-1,6-bisphosphatase (FBPase) and sedoheptulose-1,7-bisphosphatase (SBPase) are two essential activities in the Calvin-Benson-Bassham cycle that catalyze two irreversible reactions..."*
 - **PMID: 38305687:** *"The light-dependent reduction of Calvin-Benson cycle enzymes fructose 1,6-bisphosphatase (FBPase) and sedoheptulose 1,7-bisphosphatase (SBPase) was partially impaired in the FTR-knockdown ftrb mutant."*

- [PMID: 28470336](#): bisphosphatase activity → carbon assimilation (downstream — do NOT use for MF).
 - **Suggested curator questions:** (1) Is this the chloroplastic (FBP1) or cytosolic isoform? (2) Should GO:0006000 be removed project-wide from FBPase entries as a mis-generalization? (3) Should GO:0019253 (Calvin cycle) be added for FBPase entries bearing transit peptides?
 - **Suggested experiments:** recombinant FBPase substrate-specificity assay + TargetP localization to close the primary gaps.
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Proposed Follow-up Actions

1. **Apply the curation lead:** remove/demote GO:0006000; retain GO:0042132, GO:0030388, GO:0006002; add GO:0019253; scrutinize GO:0005829 and GO:0006094.
 2. **Audit sibling FBPase entries** in the AIGR corpus for the same GO:0006000 over-propagation pattern (likely a systematic TreeGraft artifact worth a batch fix).
 3. **Run orthogonal localization prediction** (TargetP2/ChloroP) and a focused phylogeny to lock the plastidic-isoform call with reproducible provenance.
 4. **Flag for experimental follow-up** the recombinant substrate-specificity assay as the single decisive test that would convert the orthology-based call into direct evidence.
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Provenance

- UniProt REST record [A0A811M8A5](#) — names, EC, catalytic activity (RHEA:11064), domain models, keywords, subcellular location, sequence.
 - QuickGO ontology REST — definitions and [is_a](#) ancestors for GO:0006000, GO:0006002, GO:0030388, GO:0042132, GO:0019318.
 - In-session logical check: GO:0006000 is NOT an [is_a](#) ancestor of GO:0006002 (False) or GO:0030388 (False); shared ancestors limited to the top-level roots.
 - In-session sequence analysis of the 411-aa protein — transit-peptide composition, FDPLDGS catalytic motif at position 182, C227–C232 CxxxxC regulatory disulfide.
 - PubMed searches on chloroplast FBPase, Calvin cycle, and thioredoxin redox regulation.
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Conclusion

NCGR_LOCUS1270 (A0A811M8A5) is a chloroplast-stromal, thioredoxin-regulated Fructose-1,6-bisphosphatase (EC 3.1.3.11) whose only reaction converts fructose-1,6-bisphosphate to fructose-6-phosphate — both phosphorylated species. Because GO:0006000 (fructose metabolic process) is defined for *free* fructose and lies on a separate ontology branch from the phospho-fructose process terms the enzyme actually performs, the GO:0006000 annotation is an **over-broad, chemically distinct over-annotation that should be removed or demoted**, retaining the more accurate MF/BP/CC and Calvin-cycle terms.

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